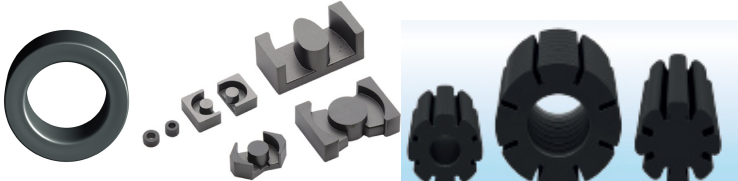


Ferrite

Material Data Sheet

Product Class:	Manganese Zinc Ferrite Cores, uncoated	
Date	31.01.18	
IMDS ID if available		
Version:	07	

Product part (IMDS: semi component)	Material Class (IMDS: Material)	Material (Classification) VDA 231	Substance	TMPS**) [wt%]	CAS if applicable	typical mass of material [wt-%]	Traces see 1)
Active Part	Ceramic	4B	Manganese Zinc Ferrite	99.5	12645-49-7	100	
			others *)	0.5			
Sum in total:						100	

Size Ø x L [max. in mm]	Part Numbers	Weight [approx. in g]
11x9 – 42.2x27	RM cores	6 – 74
50x39 – 114x93	PM cores	140 – 1940
5.7x6.15x3.9 – 21.6x24.5x15.3	EP, EPX, EPO	0.5 – 27
3.35x2.6 – 41x25	cores	0.06 – 82
14.3x8.5 – 30.5x19	P cores	3.2 – 33.3
5.3x5.25x2 – 66.4x70.5x32	TT/PR cores	0.12 – 514
7x14x5 – 20.4x53.6x50.8	E cores	1.6 – 210
5x9x6 – 37x53.5x18.3	ELP cores	0.6 – 146
16x30.6x9.8 – 31.2x59.8x22.1	ER cores	28 – 260
17.45x34.5x9.8 – 34.65x70x16.8	ETD cores	36 – 252
5.2x10.5x2.7 – 15x22.4x9.1	EC cores	0.8 – 24
9x14.8x7 – 16.4x29.7x12.5	EFD cores	5.7 – 37
92x28x93 – 108.5x30x141	EV cores	600 – 2500
	Impeders	6.5 – 3360

Not part of a product class		
Contact	Dr. Johann Reindl, MAG EPQM	Important remarks: 1) The declaration limit is 0.1% as defined by IEC 62474 (IEC PAS 61906) Traces are product parts, substances etc. that are below a percentage of 0.1 % by weight, if not otherwise regulated 2) This Material Data Sheet contains typical values of the respective products set forth herein. We expressly point out that all values and statements contained herein are based on our best present knowledge and cannot be regarded as binding statements or binding product specifications, unless otherwise explicitly agreed in writing. TDK ELECTRONICS AG AND ITS AFFILIATES HEREBY EXPRESSLY DISCLAIM ANY REPRESENTATION OR WARRANTY, WHETHER EXPRESS, IMPLIED OR STATUTORY, WITH REGARD TO THE STATEMENTS AND VALUES CONTAINED HEREIN, INCLUDING BUT NOT LIMITED TO ANY REPRESENTATION OR WARRANTY OF MERCHANTABILITY OR SUITABILITY FOR ANY PURPOSE.
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*) others: (not declarable or prohibited substances acc. GADSL) Can contain cobalt, barium, copper, silicon, vanadium, aluminum, calcium. Neither lead, cadmium, boronoxide or mercury or its compounds, nor hexavalent chromium are intentionally added **) typical mass percentage of substance		

The products set forth herein are "RoHS-compatible". RoHS-compatible means that products are compatible with the requirements according to Art. 4 (substance restrictions) of Directive 2011/65/EU of the European Parliament and of the Council of June 8th, 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

RoHS - Exemptions for the Product Class / Product according to Annex III: (☒ valid ☐ not valid)

- ☒ **no exemptions;**
- ☐ Exemption 6 (a): Lead as an alloying element in steel for machining purposes and in galvanized steel containing up to 0,35 % lead by weight;
- ☐ Exemption 6 (b): Lead as an alloying element in aluminium containing up to 0,4 % lead by weight;
- ☐ Exemption 6 (c): Copper alloy containing up to 4 % lead by weight;
- ☐ Exemption 7 (a): Lead in high melting temperature type solder (i.e. lead-based alloys containing 85 % by weight or more lead);
- ☐ Exemption 7 (c)-I: Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectric devices, or in a glass or ceramic matrix compound;
- ☐ Exemption 7 (c)-II: Lead in dielectric ceramic in capacitors for a rated voltage of 125 V AC or 250 V DC or higher;
- ☐ Exemption 7 (c)-III: Lead in dielectric ceramic in capacitors for a rated voltage of less than 125 V AC or 250 V DC;
- ☐ Exemption 15: Lead in solders to complete a viable electrical connection between semiconductor die and carrier within integrated circuit Flip Chip packages;
- ☐ Other Exemption than above